MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BVOC (HONOURS) Regular EXAMINATION April 2026

SECOND SEMESTER

Skill Development Components (SDC)- MG2SDCFTA101 – FOOD MICROBIOLOGY

(2025 ADMISSION ONWARDS)

ANSWER SCHEME

Duration : 1 Hour 30 minutes **Maximum mark : 50**

Part A

One Sentence

Answer any 10 questions

Each question carries 1 marks

1. Optical parts of microscope [K]/[CO1]
   - Eyepiece (ocular lens), objective lenses, condenser, mirror/light source
2. Types of culture media (physical state) [K]/[CO1]
   - Liquid (broth), semi-solid, solid
3. Autotrophs [K]/[CO1]
   - Organisms that synthesize their own food using CO_2 as carbon source
4. Structure for bacterial motility [K]/[CO1]
   - Flagella
5. Biosynthesis in viral replication [K]/[CO1]
   - Synthesis of viral nucleic acid and proteins inside host cell
6. Eukaryotes [K]/[CO1]
   - Organisms with true nucleus and membrane-bound organelles
7. Microbial nutrition [K]/[CO1]
   - Process by which microorganisms obtain nutrients for growth and metabolism
8. Foods associated with Ochratoxin A [U]/[CO2]
   - Cereals, coffee, dried fruits, spices
9. Organisms causing surface slime in meat [U]/[CO3]
   - Pseudomonas spp., Achromobacter
10. Microbial spoilage of food [U]/[CO3]

- Undesirable changes in food caused by microbial activity

11. Flipper [U]/[CO3]
    - A flipper can is a type of spoiled can in which both ends appear normal (flat), but when one end is pressed, the other end bulges or flips out.
12. Blue milk [U]/[CO3]
    - Bluish discoloration due to Pseudomonas syncyanea

Part B

Short Answer

Answer any 4 questions

Each questions carries 5 marks

13. Dry heat sterilization (5 marks) [K]/[CO1]

- Definition (1)
- Principle: oxidation of cell components (1)
- Methods: Flaming , hot air oven , incineration(1)
- Conditions: 160–180°C for 2 hrs (1)
- Applications: glassware, powders, oils (1)

14. Anaerobic jar components (5 marks) [K]/[CO1]

- Jar container (1)
- Gas pack (H_2 + CO_2 generation) (1)
- Catalyst (palladium) (1)
- Indicator (methylene blue/resazurin) (1)
- Function explanation (removal of O_2) (1)

15. Listeriosis (5 marks) [U]/[CO2]

- Causative agent: Listeria monocytogenes (1)
- Risk groups: pregnant, elderly, immunocompromised (1)

- Foods: milk, cheese, ready-to-eat meat (1)
- Symptoms: fever, meningitis, miscarriage (1)
- Prevention: pasteurization, hygiene (1)

16. Food poisoning (5 marks) [U]/[CO2]

- Definition (1)
- Sources: contaminated food/water (1)
- Causes: bacteria/toxins (1)
- Symptoms: vomiting, diarrhea, fever (1)
- Prevention: hygiene, proper cooking (1)

17. Giardiasis (5 marks) [U]/[CO2]

- Agent: Giardia lamblia (1)
- Transmission: contaminated water/food (1)
- Foods: raw vegetables, water (1)
- Symptoms: diarrhea, bloating (1)
- Prevention: sanitation, safe water (1)

18. Bacterial spoilage in fish (5 marks) [U]/[CO3]

- Surface slime (1)
- Off-odour (trimethylamine) (1)
- Discoloration (1)
- Softening of tissue, Autolysis (1)
- Gas production (1)

Part C

Answer any 2 questions

Each question carries 10 marks

19. Classification of fungi (10 marks) [K]/[CO1]

- Introduction (1)
- Based on morphology (3)
    - Yeasts, Yeast like, molds, dimorphic fungi
- Based on reproduction (4)
    - Asexual: spores (conidia, sporangiospores)
    - Sexual: zygospores, ascospores, basidiospores
- Examples (1–2)
- Diagram (optional)

20. Streak plate method (10 marks) [K]/[CO1]

- Definition (1)
- Principle (2)
- Procedure (3)
- Applications (2)
- Advantages (1)
- Disadvantages (1)

21. Botulism (10 marks) [K]/[CO2]

- Causative organism: Clostridium botulinum (1)
- Source: soil, canned foods (2)
- Foods involved (1)
- Incubation period (1)
- Toxin production (1)
- Types: Adult botulism and Infant botulism (1)
- Symptoms: paralysis, respiratory failure, blurred vision (2)
- Prevention: proper canning, heating (1)

22. Spoilage of fruits & vegetables (10 marks) [K]/[CO3]

- Introduction (1)
- Types of spoilage (5)
    - Fungal rot
    - Bacterial soft rot -Erwinia carotovora
    - Gray mold rot- Botrytis cineria
    - Rhizopus soft rot- Rhizopus stolonifer
    - Anthracnose- Colletotrichum lindemuthianum
    - Alternaria rot- Alternaria tenuis
    - Blue mold rot – Penicillium digitatum
    - Downey mildew- Phytophthora
    - Watery soft rot- Sclerotinia sclerotiorum
    - Stem end rots- Alternaria, fusarium
    - Black mold rot – Alternaria
    - Pink mold rot- Trichothecium roseum
    - Green mold rot- Cladosporium, Trichoderma
    - Brown rot-Sclerotinia

    - Yeast fermentation
- Causes (2)
    - Enzymatic browning
    - Physical damage
    - Microbial
- Prevention methods (2)